Math 231A HW 4

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Problem 1

Etingof Problem Sets 3.5:

(1) Prove that $\mathbb{R}^3$ with the commutator given by the cross-product is a Lie algebra. Show that this Lie algebra is isomorphic to $\mathfrak{so}(3,\mathbb{R})$.

(2) Let $\varphi : \mathfrak{so}(3,\mathbb{R}) \to \mathbb{R}^3$ be the isomorphism of part (1). Prove that under this isomorphism, the standard action of $\mathfrak{so}(3,\mathbb{R})$ on $\mathbb{R}^3$ is identified with the action of $\mathbb{R}^3$ on itself given by the cross-product:

$$a \cdot \overline{v} = \varphi(a) \times \overline{v}, \quad a \in \mathfrak{so}(3,\mathbb{R}), \overline{v} \in \mathbb{R}^3 \tag{1.1}$$

where $a \cdot \overline{v}$ is the usual multiplication of a matrix by a vector.

Solution:

(1) It's well-known that cross product is skew-symmetric, hence to check that $(\mathbb{R}^3, \times)$ forms a Lie Algebra, it remains to check Jacobi's Identity.

Recall that for standard basis $\{\overline{e}_1, \overline{e}_2, \overline{e}_3\} \subset \mathbb{R}^3$ it satisfies the cyclic relation, where $\overline{e}_1 \times \overline{e}_2 = \overline{e}_3$, $\overline{e}_2 \times \overline{e}_3 = \overline{e}_1$, and $\overline{e}_3 \times \overline{e}_1 = \overline{e}_2$. Then, given arbitrary $\overline{x} = \sum_{i=1}^3 x_i\overline{e}_i$, $\overline{y} = \sum_{i=1}^3 y_i\overline{e}_i$, and $\overline{z} = \sum_{i=1}^3 z_i\overline{e}_i$, using bilinearity of cross product, we get the following:

$$\begin{aligned}(\overline{x}\times\overline{y})\times\overline{z} &= ((x_2y_3 - x_3y_2)\overline{e}_1 + (x_3y_1 - x_1y_3)\overline{e}_2 + (x_1y_2 - x_2y_1)\overline{e}_3) \times \overline{z} \\ &= ((x_3y_1 - x_1y_3)z_3 - (x_1y_2 - x_2y_1)z_2)\overline{e}_1 \\ &\quad +((x_1y_2 - x_2y_1)z_1 - (x_2y_3 - x_3y_2)z_3)\overline{e}_2 \\ &\quad +((x_2y_3 - x_3y_2)z_2 - (x_3y_1 - x_1y_3)z_1)\overline{e}_3\end{aligned} \tag{1.2}$$

$$\begin{aligned}(\overline{y}\times\overline{z})\times\overline{x} &= ((y_3z_1 - y_1z_3)x_3 - (y_1z_2 - y_2z_1)x_2)\overline{e}_1 \\ &\quad +((y_1z_2 - y_2z_1)x_1 - (y_2z_3 - y_3z_2)x_3)\overline{e}_2 \\ &\quad +((y_2z_3 - y_3z_2)x_2 - (y_3z_1 - y_1z_3)x_2)\overline{e}_3\end{aligned} \tag{1.3}$$

$$\begin{aligned}(\overline{z}\times\overline{x})\times\overline{y} &= ((z_3x_1 - z_1x_3)y_3 - (z_1x_2 - z_2x_1)y_2)\overline{e}_1 \\ &\quad +((z_1x_2 - z_2x_1)y_1 - (z_2x_3 - z_3x_2)y_3)\overline{e}_2 \\ &\quad +((z_2x_3 - z_3x_2)y_2 - (z_3x_1 - z_1x_3)y_2)\overline{e}_3\end{aligned} \tag{1.4}$$

Which, by verifying the cross terms, $(\overline{x}\times\overline{y})\times\overline{z} + (\overline{y}\times\overline{z})\times\overline{x} + (\overline{z}\times\overline{x})\times\overline{y} = \overline{0}$, hence cross product satisfies Jacobi Identity, showing that $(\mathbb{R}^3, \times)$ forms a Lie algebra.

Now, consider the following standard basis of $\mathfrak{so}(3,\mathbb{R})$:

$$J_x := \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_y := \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad J_z := \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{1.5}$$

Notice that it satisfies the following relation using matrix commutator:

$$\begin{aligned} \left[J_x, J_y\right] &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = J_z \end{aligned} \tag{1.6}$$

$$\begin{aligned} \left[J_y, J_z\right] &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = J_x \end{aligned} \tag{1.7}$$

$$\begin{aligned} \left[J_z, J_x\right] &= \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} = J_y \end{aligned} \tag{1.8}$$

Hence, if consider the linear map $\varphi : \mathfrak{so}(3,\mathbb{R}) \to \mathbb{R}^3$ by $\varphi(J_x) = \overline{\mathrm{e}}_1$, $\varphi\left(J_y\right) = \overline{\mathrm{e}}_2$, and $\varphi(J_z) = \overline{\mathrm{e}}_3$, the following equations are satisfied:

$$\varphi\left(\left[J_x, J_y\right]\right) = \varphi(J_z) = \overline{\mathrm{e}}_3 = [\overline{\mathrm{e}}_1, \overline{\mathrm{e}}_2] = \left[\varphi(J_x), \varphi\left(J_y\right)\right] \tag{1.9}$$

$$\varphi\left(\left[J_y, J_z\right]\right) = \varphi(J_x) = \overline{\mathrm{e}}_1 = [\overline{\mathrm{e}}_2, \overline{\mathrm{e}}_3] = \left[\varphi\left(J_y\right), \varphi(J_z)\right] \tag{1.10}$$

$$\varphi([J_z, J_x]) = \varphi\left(J_y\right) = \overline{\mathrm{e}}_2 = [\overline{\mathrm{e}}_3, \overline{\mathrm{e}}_1] = [\varphi(J_z), \varphi(J_x)] \tag{1.11}$$

Hence, since φ preserves the commutators (on distinct pairs of basis) between the two Lie algebras, it forms a Lie algebra homomorphism; also, since it sends basis $\left\{J_x, J_y, J_z\right\} \subset \mathfrak{so}(3,\mathbb{R})$ to basis $\{\overline{\mathrm{e}}_1, \overline{\mathrm{e}}_2, \overline{\mathrm{e}}_3\} \subset \mathbb{R}^3$, it's in fact an $\mathbb{R}$-linear isomorphism. So, $\mathfrak{so}(3,\mathbb{R})$ and $\mathbb{R}^3$ are isomorphic Lie algebras.

(2) To verify that the natural action of $\mathfrak{so}(3,\mathbb{R})$ on $\mathbb{R}^3$ is the same as cross product action of $\mathbb{R}^3$ on itself, it suffices to verify for the basis matrices $\left\{J_x, J_y, J_z\right\} \subset \mathbb{R}^3$ (since for any $a \in \mathfrak{so}(3,\mathbb{R})$, if $a = a_1 J_x + a_2 J_y + a_3 J_z$, we have $a \cdot \overline{v} = a_1 J_x \cdot \overline{v} + a_2 J_y \cdot \overline{v} + a_3 J_z \cdot \overline{v} = a_1 \varphi(J_x) \times \overline{v} + a_2 \varphi(J_2) \times \overline{v} + a_3 \varphi(J_3) \times \overline{v} = \varphi\left(a_1 J_x + a_2 J_y + a_3 J_z\right) \times \overline{v} = \varphi(a) \times \overline{v}$).

Which, for each basis matrix of $\mathfrak{so}(3,\mathbb{R})$, we have the following for any $\overline{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3$:

$$J_x \cdot \overline{v} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ -z \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \overline{\mathrm{e}}_1 \times \overline{v} = \varphi(J_x) \times \overline{v} \tag{1.12}$$

$$J_y \cdot \overline{v} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} z \\ 0 \\ -x \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \overline{\mathrm{e}}_2 \times \overline{v} = \varphi(J_y) \times \overline{v} \tag{1.13}$$

$$J_z \cdot \overline{v} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -y \\ x \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \times \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \overline{\mathrm{e}}_3 \times \overline{v} = \varphi(J_z) \times \overline{v} \tag{1.14}$$

Hence, the $\mathfrak{so}(3,\mathbb{R})$ action on $\mathbb{R}^3$ is compatible with the cross product action of $\mathbb{R}^3$ on itself.

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Problem 2

Etingof Problem Set 3.6:

Let P_n be the space of polynomials with real coefficients of degree $, \leq n$ in variable x. The Lie group $G = \mathbb{R}$ acts on P_n by translations of the argument: $\rho(t)(x) := x + t, t \in G$. Show that the corresponding action of the Lie algebra $\mathfrak{g} = \mathbb{R}$ is given by $\rho(a) = a\partial_x, a \in \mathfrak{g}$, and deduce from this the Taylor formula for polynomials:

$$f(x+t) = \sum_{n \geq 0} \frac{(t\partial_x)^n}{n!} f \tag{2.1}$$

Solution: Here for simplicity we'll directly use $\mathbb{R}$ instead of notation G.

(1) The Exponential Map $\exp_{\mathbb{R}} : \mathfrak{g} \to \mathbb{R}$:

For every $a \in \mathfrak{g} = \mathbb{R}$, if we want the Lie group homomorphism $\gamma_a : \mathbb{R} \to \mathbb{R}$ satisfying ${\gamma_a}'(0) = a$, notice that the linear map $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = ax$ is a group homomorphism, $f'(0) = a$, and f is a smooth function (since it's a linear operator on $\mathbb{R}$), so based on the uniqueness of such Lie group Homomorphism (**Proposition 7.1** in Etingof's Notes), we have $\gamma_{a(x)} = f(x) = ax$. Hence, the exponential map $\exp_{\mathbb{R}} : \mathfrak{g} \to \mathbb{R}$ satisfies $\exp_{\mathbb{R}}(a) = \gamma_{a(1)} = a$, showing that when viewing $\mathfrak{g} = \mathbb{R}$, $\exp_{\mathbb{R}}$ is identity.

(2) The Lie algebra action of $\mathfrak{g}$ on P_n:

For all $f(x) \in P_n$, every $a \in \mathfrak{g}$ with $\exp_{\mathbb{R}}(a) = a \in \mathbb{R}$ has an action $\rho(\exp_{\mathbb{R}}(a))(f(x)) = f(x+a)$. Hence, the Lie algebra action of $a \in \mathfrak{g}$ on P_n is as follow:

$$a \cdot f(x) := \frac{d}{dt}\Big|_{t=0} \rho(\exp_{\mathbb{R}}(ta))(f(x)) = \frac{d}{dt}\Big|_{t=0} \rho(ta)(f(x))$$
$$= \frac{d}{dt}\Big|_{t=0} f(x+ta) = f'(x+ta)a\Big|_{t=0} = af'(x) = a\partial_x(f(x)) \tag{2.2}$$

So, each $a \in \mathfrak{g}$ has an action on P_n, corresponds to a linear operator $a\partial_x \in \mathrm{End}(P_n)$ (where $\mathrm{End}(P_n) = T_1(\mathrm{GL}(P_n))$, the tangent space of general linear group of P_n).

(3) The Taylor Formula:
First, notice that the froup action of $\mathbb{R}$ on P_n is in fact a linear action, or $\rho : \mathbb{R} \to \mathrm{GL}(P_n)$. Given any $a, b \in \mathbb{R}$ and $f(x), g(x) \in P_n$, we have the following for all $t \in \mathbb{R}$:

$$\rho(t)(af(x) + bg(x)) = af(x+t) + bg(x+t) = a\rho(t)(f(x)) + b\rho(t)(g(x)) \tag{2.3}$$

This shows that $\rho(t) \in \mathrm{End}(P_n)$, a linear operator on P_n. And, the reason it's invertible, is simply because $\rho(-t)$ serves as an inverse (due to the property of group action), so $\rho(t) \in \mathrm{GL}(P_n)$.

Now, if we identify $P_n \cong \mathbb{R}^{n+1}$, let $\mathfrak{k} := T_1\ \mathrm{GL}(P_n) = \mathrm{End}(P_n)$ be the tangent space of $\mathrm{GL}(P_n)$, then the exponential map $\exp_{\mathrm{GL}(P_n)} : \mathfrak{k} \to \mathrm{GL}(P_n)$ is given as operator exponential $\exp_{\mathrm{GL}(P_n)}(T) = \sum_{n=0}^{\infty} \frac{T^n}{n!}$.

Then, recall that when viewing the group action $\rho : \mathbb{R} \to \mathrm{GL}(P_n)$ as a smooth lie group homomorphism, we have $\rho \circ \exp_{\mathbb{R}} = \exp_{\mathrm{GL}(P_n)} \circ \rho_*$ (where $\rho_* : \mathfrak{g} \to \mathfrak{k}$ is the differential at $0 \in \mathbb{R}$, the identity of $\mathbb{R}$ under addition), hence for every $a \in \mathfrak{g}$, since based on the previous section we have the differential ρ_* satisfies $\rho_*(a) = a\partial_x \in \mathfrak{k} = \mathrm{End}(P_n)$ (since the previous part explicitely calculated what the corresponding tangent vector ρ_* sends a to), then we get the following formula:

$$\rho(a) = \rho \circ \exp_{\mathbb{R}}(a) = \exp_{\mathrm{GL}(P_n)} \circ \rho_*(a) = \exp_{\mathrm{GL}(P_n)}(a\partial_x) = \sum_{n=0}^{\infty} \frac{(a\partial_x)^n}{n!} \tag{2.4}$$

Hence, we derive the following Taylor's Formula (switching a with t):

$$f(x+t) = \rho(t)(x) = \sum_{n=0}^{\infty} \frac{(t\partial_x)^n}{n!} f(x) \tag{2.5}$$

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Problem 3

Etingof Problem Set 3.8:
Let $\mathrm{SL}(2, \mathbb{C})$ act on $\mathbb{CP}^1$ in the usual way:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}(x : y) = (ax + by : cx + dy) \tag{3.1}$$

This defines an action of $\mathfrak{g} = \mathfrak{sl}(2, \mathbb{C})$ by vector fields on $\mathbb{CP}^1$. Write explicitly vector fields corresponding to h, e, f in terms of coordinate $t = \frac{x}{y}$ on the open cell $\mathbb{C} \subset \mathbb{CP}^1$.

Solution:

Here, since $t = \frac{x}{y}$ is the coordinate chart, for this to make sense the open cell $\mathbb{C} \subset \mathbb{CP}^1$ corresponds to the Zariski-open set $\{y \neq 0\} \subset \mathbb{CP}^1$.

Recall that $\mathfrak{sl}(2,\mathbb{C})$ consists of all traceless matrices, which every matrix $A \in \mathfrak{sl}(2,\mathbb{C})$ is of the form $\begin{pmatrix} a & b \\ c & -a \end{pmatrix}$. Hence, $\mathfrak{sl}(2,\mathbb{C})$ has a basis of $\left\{ h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right\}$ when viewed as a complex vector space.

Now, consider the „complex curve“ $\exp(sX) : \mathbb{C} \to \mathrm{SL}(2,\mathbb{C})$ for $X = h, e, f$ and $s = \mathbb{C}$ (consider the fact that $e^2 = f^2 = 0 \in M_2(\mathbb{C})$):

$$\exp(sh) = \sum_{n=0}^{\infty} \frac{(sh)^n}{n!} = \sum_{n=0}^{\infty} \frac{\begin{pmatrix} s & 0 \\ 0 & -s \end{pmatrix}^n}{n!} = \sum_{n=0}^{\infty} \frac{1}{n!} \begin{pmatrix} s^n & 0 \\ 0 & (-s)^n \end{pmatrix} = \begin{pmatrix} e^s & 0 \\ 0 & e^{-s} \end{pmatrix} \tag{3.2}$$

$$\exp(se) = \sum_{n=0}^{\infty} \frac{(se)^n}{n!} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & s \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} \tag{3.3}$$

$$\exp(sf) = \sum_{n=0}^{\infty} \frac{(sf)^n}{n!} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ s & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix} \tag{3.4}$$

Which, if express in the coordinate $t = \frac{x}{y}$ on the cell $\mathbb{C} \coloneqq \{y \neq 0\} \subset \mathbb{CP}^1$, we get the following for all $t = \frac{x}{y} \in \mathbb{C}$ (if let $v(X)$ denotes the vector field corresponding to $X = h, e, f$):

(Note: Here we abused a bit of notation, letting $t = \frac{x}{y}$ and $(x : y)$ be interchangeable).

$$\begin{aligned} v(h)(t) &= \frac{d}{ds}\bigg|_{s=0} \exp(sh)(x:y) = \frac{d}{ds}\bigg|_{s=0} \begin{pmatrix} e^s & 0 \\ 0 & e^{-s} \end{pmatrix}(x:y) = \frac{d}{ds}\bigg|_{s=0} (e^s x : e^{-s} y) \\ &= \frac{d}{ds}\bigg|_{s=0} \frac{e^s x}{e^{-s} y} = \frac{d}{ds}\bigg|_{s=0} e^{2s} t = 2t \end{aligned} \tag{3.5}$$

$$\begin{aligned} v(e)(t) &= \frac{d}{ds}\bigg|_{s=0} \exp(se)(x:y) = \frac{d}{ds}\bigg|_{s=0} \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}(x:y) = \frac{d}{ds}\bigg|_{s=0} (x + sy : y) \\ &= \frac{d}{ds}\bigg|_{s=0} \frac{x + sy}{y} = \frac{d}{ds}\bigg|_{s=0} t + s = 1 \end{aligned} \tag{3.6}$$

$$\begin{aligned} v(f)(t) &= \frac{d}{ds}\bigg|_{s=0} \exp(sf)(x:y) = \frac{d}{ds}\bigg|_{s=0} \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}(x:y) = \frac{d}{ds}\bigg|_{s=0} (x : sx + y) \\ &= \frac{d}{ds}\bigg|_{s=0} \frac{x}{sx + y} = -\frac{x^2}{(sx+y)^2}\bigg|_{s=0} = -\frac{x^2}{y^2} = -t^2 \end{aligned} \tag{3.7}$$

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Problem 4

Etingof Problem Set 3.9:

Let G be a Lie group with Lie algebra $\mathfrak{g}$, $\mathrm{Aut}(\mathfrak{g})$ the group of automorphisms of $\mathfrak{g}$, and $\mathrm{Der}(\mathfrak{g})$ be the Lie algebra of derivations of $\mathfrak{g}$.

(1) Show that $g \mapsto \mathrm{Ad}\ g$ gives a morphism of Lie groups $G \to \mathrm{Aut}(\mathfrak{g})$; similarly, $x \mapsto \mathrm{ad}\ x$ is a morphism of of Lie algebras $\mathfrak{g} \to \mathrm{Der}(\mathfrak{g})$ (The automorphisms of the form $\mathrm{Ad}\ g$ are called *inner automorphisms* the derivations of the form $\mathrm{ad}\ x, x \in \mathfrak{g}$ are called *inner derivations*).

(2) Show that for $f \in \mathrm{Der}(\mathfrak{g})$, $x \in \mathfrak{g}$, one has $[f, \mathrm{ad}\ x] = \mathrm{ad}\ f(x)$ as operators in $\mathfrak{g}$, and deduce from this that $\mathrm{ad}(\mathfrak{g})$ is an ideal in $\mathrm{Der}(\mathfrak{g})$.

Solution:

(1) First, given the map $G \to \mathrm{Aut}(\mathfrak{g})$ by $g \mapsto \mathrm{Ad}\ g$ (where $\mathrm{Ad}\ g \in \mathrm{Aut}(\mathfrak{g})$ is formally defined as $\mathrm{Ad}(g)_* : \mathfrak{g} \to \mathfrak{g}$, where $\mathfrak{g} = T_1 G$, the tangent space at identity). Which, since given any $g, h, k \in G$, we have the following formula:

$$\mathrm{Ad}(gh)(k) = (gh)k(gh)^{-1} = (gh)k(h^{-1}g^{-1}) = \mathrm{Ad}(g)(hkh^{-1}) = \mathrm{Ad}(g)(\mathrm{Ad}(h)(k)) \tag{4.1}$$

Then, we get that $\mathrm{Ad}(gh) = \mathrm{Ad}(g) \circ \mathrm{Ad}(h)$, which implies that $\mathrm{Ad}(gh)_* = \mathrm{Ad}(g)_* \circ \mathrm{Ad}(h)_*$ (since here everything is evaluated at $1 \in G$, which is invariant under any adjoint action on the Lie group itself). So, we conclude that $gh \mapsto \mathrm{Ad}\ g \circ \mathrm{Ad}\ h \in \mathrm{Aut}(\mathfrak{g})$, showing that it's indeed a group homomorphism.

The reason why this is a Lie group homomorphism, is because as $g \in G$ varies smoothly, defining a local chart near 1 realizes a basis on $\mathfrak{g} = T_1 G$, which then identifies $\mathrm{GL}(\mathfrak{g})$ as $\mathrm{GL}_{n(\mathbb{R})}$ (so it has a natural smooth manifold structure). Then, with g varies smoothly, the corresponding differential $\mathrm{Ad}(g)_*$ written in terms of the basis of $\mathfrak{g}$, also has entries that variy smoothly, hence $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$ is smooth.

Now, given the other map $\mathfrak{g} \to \mathrm{Der}(\mathfrak{g})$ by $x \mapsto \mathrm{ad}\ x$. To verify that it's well-defined, recall that $\mathrm{ad}\ x(y) := [x, y]$ for all $x, y \in \mathfrak{g}$, then by Jacobi Identity, we have the following:

$$\begin{aligned} \mathrm{ad}\ x([y,z]) &= [x,[y,z]] = -[[y,z],x] = [[x,y],z] + [[z,x],y] = [\mathrm{ad}\ x(y), z] - [[x,z],y] \\ &= [\mathrm{ad}\ x(y), z] + [y,[x,z]] = [\mathrm{ad}\ x(y), z] + [y, \mathrm{ad}\ x(z)] \end{aligned} \tag{4.2}$$

Hence, $\mathrm{ad}\ x$ is indeed a derivation on $\mathfrak{g}$, so the map is well-defined.

Then, to verify that it's a linear map, given any $a, b \in \mathbb{K}$ (the base field of $\mathfrak{g}$) and $x, y, z \in \mathfrak{g}$, we have the following:

$$\mathrm{ad}\ (ax + by)(z) = [ax + by, z] = a[x,z] + b[y,z] = a\ \mathrm{ad}(x)(z) + b\ \mathrm{ad}(y)(z) \tag{4.3}$$

This shows that $\mathrm{ad}(ax + by) = a\ \mathrm{ad}(x) + b\ \mathrm{ad}(y)$, which ad is a linear map.

Finally, to show it's a Lie algebra homomorphism, we again utilize the Jacobi Identity. For every $x, y, z \in \mathfrak{g}$, we have the following:

$$\begin{aligned} \mathrm{ad}([x,y])(z) &= [[x,y],z] = -[[y,z],x] - [[z,x],y] = [x,[y,z]] + [y,[z,x]] \\ &= [x, \mathrm{ad}\ y(z)] - [y,[x,z]] = \mathrm{ad}\ x(\mathrm{ad}\ y(z)) - [y, \mathrm{ad}\ x(z)] \\ &= (\mathrm{ad}\ x) \circ (\mathrm{ad}\ y)(z) - (\mathrm{ad}\ y) \circ (\mathrm{ad}\ x)(z) = [\mathrm{ad}\ x, \mathrm{ad}\ y](z) \end{aligned} \tag{4.4}$$

(Note: Here if viewing $\mathrm{Der}(\mathfrak{g}) \subseteq \mathrm{End}(\mathfrak{g})$ as a Lie subalgebra, its Lie Bracket is given by the commutator of the operators).

Hence, we get that $\mathrm{ad}([x,y]) = [\mathrm{ad}\ x, \mathrm{ad}\ y]$, showing that ad is indeed a Lie algebra homomorphism.

(2) Given any $f \in \mathrm{Der}(\mathfrak{g})$, as a derivation, for any $x, y \in \mathfrak{g}$, it satisfies the following formula:

$$f([x,y]) = [f(x), y] + [x, f(y)] \tag{4.5}$$

So, given any $x, y, z \in \mathfrak{g}$, we have the following computation:

$$\begin{aligned} [f, \mathrm{ad}\ x](y) &= f \circ (\mathrm{ad}\ x)(y) - (\mathrm{ad}\ x) \circ f(y) = f([x,y]) - [x, f(y)] \\ &= [f(x), y] + [x, f(y)] - [x, f(y)] = [f(x), y] = \mathrm{ad}\ f(x)(y) \end{aligned} \tag{4.6}$$

Hence, this verifies that $[f, \mathrm{ad}\ x] = \mathrm{ad}\ f(x)$, showing that $\mathrm{ad}(\mathfrak{g}) \subseteq \mathrm{Der}(\mathfrak{g})$ is an ideal (since given any $f \in \mathrm{Der}(\mathfrak{g})$ and $\mathrm{ad}\ x \in \mathrm{ad}(\mathfrak{g})$, it follows that $[f, \mathrm{ad}\ x] = \mathrm{ad}\ f(x) \in \mathrm{ad}(\mathfrak{g})$).

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Problem 5

Etingof Problem Set 3.11:

Let J_x, J_y, J_z be the standard basis in $\mathfrak{so}(3,\mathbb{R}) \cong \mathbb{R}^3$ (the Lie bracket is the cross product). The standard action of $\mathrm{SO}(3,\mathbb{R})$ on $\mathbb{R}^3$ defines an action of $\mathfrak{so}(3,\mathbb{R})$ by vector fields on $\mathbb{R}^3$. Abusing the language, we will use the same notation J_x, J_y, J_z for the corresponding vector fields on $\mathbb{R}^3$. Let $\Delta_{\text{sph}} = J_x^2 + J_y^2 + J_z^2$; this is a second order differential operator on $\mathbb{R}^3$, which is usually called the *Spherical Laplace Operator*, or the *Laplace Operator on the Sphere.*

(1) Write Δ_{sph} in terms of $x, y, z, \partial_x, \partial_y, \partial_z$.

(2) Show that Δ_{sph} is well-defined as a differential operator on a sphere $S^2 = \{(x,y,z) | x^2 + y^2 + z^2 = 1\}$, i.e. if f is a function on $\mathbb{R}^3$ then $(\Delta_{\text{sph}} f)|_{S^2}$ only depends on $F|_{S^2}$.

(3) Show that the usual Laplace operator $\Delta = \partial_x^2 + \partial_y^2 + \partial_z^2$ can be written in the form $\Delta = \frac{1}{r^2}\Delta_{\text{sph}} + \Delta_{\text{radial}}$, where Δ_{radial} is a differential operator written in terms of $r = \sqrt{x^2+y^2+z^2}$ and $r\partial_r = x\partial_x + y\partial_y + z\partial_z$.

(4) Show that Δ_{sph} is rotation invariant: For any function f (on $\mathbb{R}^3$) and $g \in \mathrm{SO}(3,\mathbb{R})$, $\Delta_{\text{sph}}(gf) = g(\Delta_{\text{sph}} f)$.

Solution:

(1) First, we need to compute what the vector fields are. For any $u \in \mathbb{R}^3$, we'll identify its tangent space $T_u\mathbb{R}^3 := \mathrm{span}\{\partial_x, \partial_y, \partial_z\}$, where each ∂_i corresponds to the direction i geometrically (since it's being interpreted as directional derivatives when the space lies in $\mathbb{R}^n$, in this case it's $\mathbb{R}^3$ itself).

For simplicity, we'll first calculate the curve $\exp(tJ_i) : \mathbb{R} \to \mathrm{SO}(3,\mathbb{R})$ for each J_i. Which, given that $J_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$, $J_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}$, $J_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, they satisfy the identities $J_x^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$, $J_y^2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$, $J_z^2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. So, their corresponding curve $\exp(tJ_i) : \mathbb{R} \to \mathrm{SO}(3,\mathbb{R})$ for each J_i is given as follow:

$$\exp(tJ_x) = \sum_{n=0}^{\infty} \frac{(tJ_x)^n}{n!} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \sum_{k=1}^{\infty} t^{2k} \frac{J_x^{2k}}{(2k)!} + t^{2k-1} \frac{J_x^{2k-1}}{(2k-1)!}$$
$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \sum_{k=1}^{\infty} t^{2k} \frac{\begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}^k}{(2k)!} + t^{2k-1} \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}^k \frac{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}}{(2k-1)!}$$
$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \sum_{k=0}^{\infty} t^{2k} \frac{\begin{pmatrix} 0 & 0 & 0 \\ 0 & (-1)^k & 0 \\ 0 & 0 & (-1)^k \end{pmatrix}}{(2k)!} + t^{2k+1} \frac{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & (-1)^k \\ 0 & -(-1)^k & 0 \end{pmatrix}}{(2k+1)!} \tag{5.1}$$
$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & \cos(t) & 0 \\ 0 & 0 & \cos(t) \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & \sin(t) \\ 0 & -\sin(t) & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(t) & \sin(t) \\ 0 & -\sin(t) & \cos(t) \end{pmatrix}$$

$$\exp(tJ_y) = \sum_{n=0}^{\infty} \frac{(tJ_y)^n}{n!} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \sum_{k=1}^{\infty} t^{2k} \frac{J_y^{2k}}{(2k)!} + t^{2k-1} \frac{J_y^{2k-1}}{(2k-1)!}$$
$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \sum_{k=1}^{\infty} t^{2k} \frac{\begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}^k}{(2k)!} + t^{2k-1} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}^k \frac{\begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}}{(2k-1)!}$$
$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \sum_{k=0}^{\infty} t^{2k} \frac{\begin{pmatrix} (-1)^k & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & (-1)^k \end{pmatrix}}{(2k)!} + t^{2k+1} \frac{\begin{pmatrix} 0 & 0 & (-1)^k \\ 0 & 0 & 0 \\ -(-1)^k & 0 & 0 \end{pmatrix}}{(2k+1)!} \tag{5.2}$$
$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} \cos(t) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \cos(t) \end{pmatrix} + \begin{pmatrix} 0 & 0 & \sin(t) \\ 0 & 0 & 0 \\ -\sin(t) & 0 & 0 \end{pmatrix} = \begin{pmatrix} \cos(t) & 0 & \sin(t) \\ 0 & 1 & 0 \\ -\sin(t) & 0 & \cos(t) \end{pmatrix}$$

$$\exp(tJ_z) = \sum_{n=0}^{\infty} \frac{(tJ_z)^n}{n!} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \sum_{k=1}^{\infty} t^{2k} \frac{J_z^{2k}}{(2k)!} + t^{2k-1} \frac{J_z^{2k-1}}{(2k-1)!}$$
$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \sum_{k=1}^{\infty} t^{2k} \frac{\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}^k}{(2k)!} + t^{2k-1} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}^k \frac{\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}{(2k-1)!}$$
$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \sum_{k=0}^{\infty} t^{2k} \frac{\begin{pmatrix} (-1)^k & 0 & 0 \\ 0 & (-1)^k & 0 \\ 0 & 0 & 0 \end{pmatrix}}{(2k)!} + t^{2k-1} \frac{\begin{pmatrix} 0 & -(-1)^k & 0 \\ (-1)^k & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}{(2k+1)!} \tag{5.3}$$
$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} \cos(t) & 0 & 0 \\ 0 & \cos(t) & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & -\sin(t) & 0 \\ \sin(t) & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} \cos(t) & -\sin(t) & 0 \\ \sin(t) & \cos(t) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Which, $\exp(tJ_i)$ in fact corresponds to the elementary rotation matrix around the i-axis, with angle $\pm t$ (depending on the orientation).

So, if write down the corresponding vector field (using $v(J_i)$ as vector field of J_i for now), we get the following for all $u = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3$:

$$\begin{aligned} v(J_x)(u) &= \frac{d}{dt}\Big|_{t=0} \exp(tJ_x)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{d}{dt}\Big|_{t=0} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(t) & \sin(t) \\ 0 & -\sin(t) & \cos(t) \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} \\ &= \frac{d}{dt}\Big|_{t=0} \begin{pmatrix} x \\ \cos(t)y + \sin(t)z \\ -\sin(t)y + \cos(t)z \end{pmatrix} = \begin{pmatrix} 0 \\ -\sin(t)y + \cos(t)z \\ -\cos(t)y - \sin(t)z \end{pmatrix}\Big|_{t=0} = \begin{pmatrix} 0 \\ z \\ -y \end{pmatrix} = z\partial_y - y\partial_z \end{aligned} \tag{5.4}$$

$$\begin{aligned} v(J_y)(u) &= \frac{d}{dt}\Big|_{t=0} \exp(tJ_y)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{d}{dt}\Big|_{t=0} \begin{pmatrix} \cos(t) & 0 & \sin(t) \\ 0 & 1 & 0 \\ -\sin(t) & 0 & \cos(t) \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} \\ &= \frac{d}{dt}\Big|_{t=0} \begin{pmatrix} \cos(t)x + \sin(t)z \\ y \\ -\sin(t)x + \cos(t)z \end{pmatrix} = \begin{pmatrix} -\sin(t)x + \cos(t)z \\ 0 \\ -\cos(t)x - \sin(t)z \end{pmatrix}\Big|_{t=0} = \begin{pmatrix} z \\ 0 \\ -x \end{pmatrix} = z\partial_x - x\partial_z \end{aligned} \tag{5.5}$$

$$\begin{aligned} v(J_z)(u) &= \frac{d}{dt}\Big|_{t=0} \exp(tJ_z)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{d}{dt}\Big|_{t=0} \begin{pmatrix} \cos(t) & -\sin(t) & 0 \\ \sin(t) & \cos(t) & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} \\ &= \frac{d}{dt}\Big|_{t=0} \begin{pmatrix} \cos(t)x - \sin(t)y \\ \sin(t)x + \cos(t)y \\ z \end{pmatrix} = \begin{pmatrix} -\sin(t)x - \cos(t)y \\ \cos(t)x - \sin(t)y \\ 0 \end{pmatrix}\Big|_{t=0} = \begin{pmatrix} -y \\ x \\ 0 \end{pmatrix} = -y\partial_x + x\partial_y \end{aligned} \tag{5.6}$$

If abuse the notation and use J_i instead of $v(J_i)$ as the vector field, notice that since each $J_i^2 = j\partial_k - k\partial_j$ given that variables $i, j, k \in \{x, y, z\}$ are distinct, then we can conclude the following formula for every $f \in C^\infty(\mathbb{R}^3)$:

$$\begin{aligned} J_i^2(f) &= \big(j\partial_k - k\partial_j\big)\big(j \cdot \partial_k f - k \cdot \partial_j f\big) \\ &= j^2(\partial_k^2 f) - j\big(\partial_j f\big) - jk\big(\partial_k\partial_j f\big) - k(\partial_k f) - jk\big(\partial_j\partial_k f\big) + k^2\big(\partial_j^2 f\big) \end{aligned} \tag{5.7}$$

Hence, as a formula we get $J_i^2 = j^2\partial_k^2 + k^2\partial_j^2 - 2jk\partial_j\partial_k - j\partial_j - k\partial_k$. So, the Spherical Laplace Operator becomes:

$$\begin{aligned} \Delta_{\text{sph}} = J_x^2 + J_y^2 + J_z^2 &= y^2\partial_z^2 + z^2\partial_y^2 - 2yz\partial_y\partial_z - y\partial_y - z\partial_z \\ &\quad + z^2\partial_x^2 + x^2\partial_z^2 - 2xz\partial_x\partial_z - x\partial_x - z\partial_z \\ &\quad + x^2\partial_y^2 + y^2\partial_x^2 - 2xy\partial_x\partial_y - x\partial_x - y\partial_y \\ &= (y^2 + z^2)\partial_x^2 + (x^2 + z^2)\partial_y^2 + (x^2 + y^2)\partial_z^2 \\ &\quad - 2\big(xy\partial_x\partial_y + xz\partial_x\partial_z + yz\partial_y\partial_z\big) \\ &\quad - 2\big(x\partial_x + y\partial_y + z\partial_z\big) \end{aligned} \tag{5.8}$$

(2) Geometrically, given any $u \in S^2 \subset \mathbb{R}^3$, its geometric tangent space (with coordinates in $\mathbb{R}^3$) can be characterized by the plane $\{u\}^\perp = \{v \in \mathbb{R}^3 \mid u \cdot v = 0\}$. Which, notice

that if $u = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, then the vector field (expressed as geometric tangent vectors) have $J_{x(u)} = \begin{pmatrix} 0 \\ z \\ -y \end{pmatrix}, J_{y(u)} = \begin{pmatrix} z \\ 0 \\ -x \end{pmatrix}, J_{z(u)} = \begin{pmatrix} -y \\ x \\ 0 \end{pmatrix}$ at the point u, which one can verify that $J_{x(u)}, J_{y(u)}, J_{z(u)} \in \{u\}^{\perp}$, which all lie in the tangent space of $u \in S^2$. Hence, if considering each J_i as differential operator, we have $J_i|_{S^2}$ (the vector field / differential operator restricting to functions with domain S^2) being a well-defined and coincide with the original operator's property, hence $\Delta_{\text{sph}} f|_{S^2}$ being only dependent on $f|_{S^2}$ (since each $J_i f|_{S^2} = J_i|_{S^2}\ (f|_{S^2})$).

(3) If consider the general Laplace Operator $\Delta = \partial_x^2 + \partial_y^2 + \partial_z^2$, if consider the expression $\Delta - \Delta_{\text{sph}}$, together with $r = \sqrt{x^2+y^2+z^2}$ and $r\partial_r := x\partial_x + y\partial_y + z\partial_z$, we get:

$$\begin{aligned} \Delta - \frac{1}{r^2}\Delta_{\text{sph}} &= \partial_x^2 + \partial_y^2 + \partial_z^2 - \frac{y^2+z^2}{r^2}\partial_x^2 - \frac{x^2+z^2}{r^2}\partial_y^2 - \frac{x^2+y^2}{r^2}\partial_z^2 \\ &\quad + \frac{2}{r^2}\big(xy\partial_x\partial_y + xz\partial_x\partial_z + yz\partial_y\partial_z\big) + \frac{2}{r^2}\big(x\partial_x + y\partial_y + z\partial_z\big) \\ &= \frac{1}{r^2}\big(x^2\partial_x^2 + y^2\partial_y^2 + z^2\partial_z^2\big) + \frac{2}{r^2}\big(xy\partial_x\partial_y + xz\partial_x\partial_z + yz\partial_y\partial_z\big) \\ &\quad + \frac{2}{r^2}\big(x\partial_x + y\partial_y + z\partial_z\big) \end{aligned} \tag{5.9}$$

Which, notice that if consider the expression $(r\partial_r)^2$, we get the following for all $f \in C^\infty(\mathbb{R}^3)$:

$$\begin{aligned} (r\partial_r)^2(f) &= \big(x\partial_x + y\partial_y + z\partial_z\big)\big(x(\partial_x f) + y(\partial_y f) + z(\partial_z f)\big) \\ &= x(\partial_x f) + x^2(\partial_x^2 f) + xy(\partial_x\partial_y f) + xz(\partial_x\partial_z f) \\ &\quad + yx(\partial_y\partial_x f) + y(\partial_y f) + y^2(\partial_y^2 f) + yz(\partial_y\partial_z f) \\ &\quad + zx(\partial_z\partial_x f) + zy(\partial_z\partial_y f) + z(\partial_z f) + z^2(\partial_z^2 f) \\ &= \big(x^2\partial_x^2 + y^2\partial_y^2 + z^2\partial_z^2\big)(f) + 2\big(xy\partial_x\partial_y + xz\partial_x\partial_z + yz\partial_y\partial_z\big)(f) \\ &\quad + \big(x\partial_x + y\partial_y + z\partial_z\big)(f) \end{aligned} \tag{5.10}$$

So, we conclude that $(r\partial_r)^2 = \big(x^2\partial_x^2 + y^2\partial_y^2 + z^2\partial_z^2\big) + 2\big(xy\partial_x\partial_y + xz\partial_x\partial_z + yz\partial_y\partial_z\big) + \big(x\partial_x + y\partial_y + z\partial_z\big)$. So, as conclusion the operator $\Delta - \frac{1}{r^2}\Delta_{\text{sph}}$ can be written as:

$$\begin{aligned} \Delta - \frac{1}{r^2}\Delta_{\text{sph}} &= \frac{1}{r^2}\big(x^2\partial_x^2 + y^2\partial_y^2 + z^2\partial_z^2\big) + \frac{2}{r^2}\big(xy\partial_x\partial_y + xz\partial_x\partial_z + yz\partial_y\partial_z\big) \\ &\quad + \frac{2}{r^2}\big(x\partial_x + y\partial_y + z\partial_z\big) \\ &= \frac{1}{r^2}\Big(\big(x^2\partial_x^2 + y^2\partial_y^2 + z^2\partial_z^2\big) + 2\big(xy\partial_x\partial_y + xz\partial_x\partial_z + yz\partial_y\partial_z\big) + \big(x\partial_x + y\partial_y + z\partial_z\big)\Big) \\ &\quad + \frac{1}{r^2}\big(x\partial_x + y\partial_y + z\partial_z\big) \\ &= \frac{1}{r^2}(r\partial_r)^2 + \frac{1}{r^2}(r\partial_r) = \frac{1}{r}\partial_r(r\partial_r) + \frac{1}{r}\partial_r = \frac{2}{r}\partial_r + \partial_r^2 \end{aligned} \tag{5.11}$$

So, if define $\Delta_{\text{radial}} := \Delta - \frac{1}{r^2}\Delta_{\text{sph}} = \partial_r^2 + \frac{2}{r}\partial_r$, one yields $\Delta = \Delta_{\text{sph}} + \Delta_{\text{radial}}$.

(4) Before proving the rotation invariance, first recall that every $R \in \mathrm{SO}(3, \mathbb{R})$ can be decomposed into products of elementary rotation matrices with some angles (around the x, y, z -axes respectively), which are given by $R = \exp(\alpha J_x)\exp(\beta J_y)\exp(\gamma J_z)$ for some $\alpha, \beta, \gamma \in \mathbb{R}$. So, to prove such rotation invariance, it suffices to prove it for the mentioned elementary rotation matrices above.

For definiteness, we'll demonstrate it for the case of $\exp(\gamma J_z)$ (since the other two are similar, just up to some swap in coordinates and signs). Given that $(Rf)\begin{pmatrix} x \\ y \\ z \end{pmatrix} := f\left(R\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right)$ for any $R \in \mathrm{SO}(3, \mathbb{R})$, then $\exp(\gamma J_z)(\Delta_{\mathrm{sph}} f)$ is given as follow:

$$\begin{aligned}
\big(\exp(\gamma J_z)(\Delta_{\mathrm{sph}} f)\big)\begin{pmatrix} x \\ y \\ z \end{pmatrix} &= (\Delta_{\mathrm{sph}} f)\left(\begin{pmatrix} \cos(\gamma) & -\sin(\gamma) & 0 \\ \sin(\gamma) & \cos(\gamma) & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) \\
&= (\Delta_{\mathrm{sph}} f)\begin{pmatrix} \cos(\gamma)x - \sin(\gamma)y \\ \sin(\gamma)x + \cos(\gamma)y \\ z \end{pmatrix} \\
&= (\sin^2(\gamma)x^2 + \cos^2(\gamma)y^2 + 2\sin(\gamma)\cos(\gamma)xy + z^2)(\partial_x^2 f) \\
&\quad +(\cos^2(\gamma)x^2 + \sin^2(\gamma)y^2 - 2\sin(\gamma)\cos(\gamma)xy + z^2)(\partial_y^2 f) \\
&\quad +(x^2 + y^2)(\partial_z^2 f) \\
&\quad -2\big(\sin(\gamma)\cos(\gamma)(x^2 - y^2) + (\cos^2(\gamma) - \sin^2(\gamma))xy\big)(\partial_x\partial_y f) \\
&\quad -2(\cos(\gamma)xz - \sin(\gamma)yz)(\partial_x\partial_z f) - 2(\sin(\gamma)xz + \cos(\gamma)yz)(\partial_y\partial_z f) \\
&\quad -2(\cos(\gamma)x - \sin(\gamma)y)(\partial_x f) - 2(\sin(\gamma)x + \cos(\gamma)y)(\partial_y f) - z(\partial_z f)
\end{aligned} \tag{5.12}$$

Now, for the other expression $\Delta_{\mathrm{sph}}(\exp(\gamma J_z)f)$, first we'll consider the second order differential operator's effect on $\exp(\gamma J_z)f$ (Note: the first order expression are included in the second order ones):

$$\begin{aligned}
\partial_x^2\left[(\exp(\gamma J_z)f)\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right] &= \partial_x^2\left[f\left(\begin{pmatrix} \cos(\gamma) & -\sin(\gamma) & 0 \\ \sin(\gamma) & \cos(\gamma) & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right)\right] = \partial_x^2\left[f\begin{pmatrix} \cos(\gamma)x - \sin(\gamma)y \\ \sin(\gamma)x + \cos(\gamma)y \\ z \end{pmatrix}\right] \\
&= \partial_x\left[\big(\cos(\gamma)(\partial_x f) + \sin(\gamma)(\partial_y f)\big)\begin{pmatrix} \cos(\gamma)x - \sin(\gamma)y \\ \sin(\gamma)x + \cos(\gamma)y \\ z \end{pmatrix}\right] \\
&= \cos^2(\gamma)(\partial_x^2 f) + 2\sin(\gamma)\cos(\gamma)(\partial_x\partial_y f) + \sin^2(\gamma)(\partial_y^2 f)
\end{aligned} \tag{5.13}$$

$$\begin{aligned}
\partial_y\partial_x\left[(\exp(\gamma J_z)f)\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right] &= \partial_y\partial_x\left[f\begin{pmatrix} \cos(\gamma)x - \sin(\gamma)y \\ \sin(\gamma)x + \cos(\gamma)y \\ z \end{pmatrix}\right] \\
&= \partial_y\left[\big(\cos(\gamma)(\partial_x f) + \sin(\gamma)(\partial_y f)\big)\begin{pmatrix} \cos(\gamma)x - \sin(\gamma)y \\ \sin(\gamma)x + \cos(\gamma)y \\ z \end{pmatrix}\right] \\
&= \sin(\gamma)\cos(\gamma)(\partial_y^2 f - \partial_x^2 f) + (\cos^2(\gamma) - \sin^2(\gamma))(\partial_y\partial_x f)
\end{aligned} \tag{5.14}$$

$$
\begin{aligned}
\partial_z\partial_x\left[(\exp(\gamma J_z)f)\begin{pmatrix}x\\y\\z\end{pmatrix}\right] &= \partial_z\partial_x\left[f\begin{pmatrix}\cos(\gamma)x-\sin(\gamma)y\\ \sin(\gamma)x+\cos(\gamma)y\\ z\end{pmatrix}\right]\\
&= \partial_z\left[\big(\cos(\gamma)(\partial_x f)+\sin(\gamma)\big(\partial_y f\big)\big)\begin{pmatrix}\cos(\gamma)x-\sin(\gamma)y\\ \sin(\gamma)x+\cos(\gamma)y\\ z\end{pmatrix}\right]\\
&= \cos(\gamma)(\partial_z\partial_x f)+\sin(\gamma)\big(\partial_z\partial_y f\big)
\end{aligned} \tag{5.15}
$$

$$
\begin{aligned}
\partial_y^2\left[(\exp(\gamma J_z)f)\begin{pmatrix}x\\y\\z\end{pmatrix}\right] &= \partial_y^2\left[f\begin{pmatrix}\cos(\gamma)x-\sin(\gamma)y\\ \sin(\gamma)x+\cos(\gamma)y\\ z\end{pmatrix}\right]\\
&= \partial_y\left[\big(-\sin(\gamma)(\partial_x f)+\cos(\gamma)\big(\partial_y f\big)\big)\begin{pmatrix}\cos(\gamma)x-\sin(\gamma)y\\ \sin(\gamma)x+\cos(\gamma)y\\ z\end{pmatrix}\right]\\
&= \sin^2(\gamma)\big(\partial_x^2 f\big)-2\sin(\gamma)\cos(\gamma)\big(\partial_x\partial_y f\big)+\cos^2(\gamma)\big(\partial_y^2 f\big)
\end{aligned} \tag{5.16}
$$

$$
\begin{aligned}
\partial_z\partial_y\left[(\exp(\gamma J_z)f)\begin{pmatrix}x\\y\\z\end{pmatrix}\right] &= \partial_z\partial_y\left[f\begin{pmatrix}\cos(\gamma)x-\sin(\gamma)y\\ \sin(\gamma)x+\cos(\gamma)y\\ z\end{pmatrix}\right]\\
&= \partial_z\left[\big(-\sin(\gamma)(\partial_x f)+\cos(\gamma)\big(\partial_y f\big)\big)\begin{pmatrix}\cos(\gamma)x-\sin(\gamma)y\\ \sin(\gamma)x+\cos(\gamma)y\\ z\end{pmatrix}\right]\\
&= -\sin(\gamma)(\partial_z\partial_x f)+\cos(\gamma)\big(\partial_z\partial_y f\big)
\end{aligned} \tag{5.17}
$$

$$
\partial_z^2\left[(\exp(\gamma J_z)f)\begin{pmatrix}x\\y\\z\end{pmatrix}\right] = \partial_z^2\left[f\begin{pmatrix}\cos(\gamma)x-\sin(\gamma)y\\ \sin(\gamma)x+\cos(\gamma)y\\ z\end{pmatrix}\right] = \big(\partial_z^2 f\big) \tag{5.18}
$$

(Note: The final expression indicates doing the differential operators first, then evaluate with $\exp(\gamma J_z)\begin{pmatrix}x\\y\\z\end{pmatrix}$. Because there's no enough space we'll temporarily ignore the input).

Hence, if consider $g \coloneqq \exp(\gamma J_z)f$, we get the following for $\Delta_{\text{sph}}g$:

$$
\begin{aligned}
\Delta_{\text{sph}} g = {} & (y^2+z^2)(\partial_x^2 g) + (x^2+z^2)\big(\partial_y^2 g\big) + (x^2+y^2)(\partial_z^2 g) \\
& -2xy\big(\partial_x\partial_y g\big) - 2xz(\partial_x\partial_z g) - 2yz\big(\partial_y\partial_z g\big) \\
& -2x(\partial_x g) - 2y\big(\partial_y g\big) - 2z(\partial_z g) \\
= {} & (y^2+z^2)\big(\cos^2(\gamma)(\partial_x^2 f) + 2\sin(\gamma)\cos(\gamma)\big(\partial_x\partial_y f\big) + \sin^2(\gamma)\big(\partial_y^2 f\big)\big) \\
& +(x^2+z^2)\big(\sin^2(\gamma)(\partial_x^2 f) - 2\sin(\gamma)\cos(\gamma)\big(\partial_x\partial_y f\big) + \cos^2(\gamma)\big(\partial_y^2 f\big)\big) \\
& +(x^2+y^2)(\partial_z^2 f) - 2xy\big(\sin(\gamma)\cos(\gamma)\big(\partial_y^2 f - \partial_x^2 f\big) + \big(\cos^2(\gamma) - \sin^2(\gamma)\big)\big(\partial_y\partial_x f\big)\big) \\
& -2xz\big(\cos(\gamma)(\partial_z\partial_x f) + \sin(\gamma)\big(\partial_z\partial_y f\big)\big) - 2yz\big(-\sin(\gamma)(\partial_z\partial_x f) + \cos(\gamma)\big(\partial_z\partial_y f\big)\big) \\
& -2x\big(\cos(\gamma)(\partial_x f) + \sin(\gamma)\big(\partial_y f\big)\big) - 2y\big(-\sin(\gamma)(\partial_x f) + \cos(\gamma)\big(\partial_y f\big)\big) - 2z(\partial_z f)
\end{aligned}
\tag{5.19}
$$

$$
\begin{aligned}
= {} & \big(\sin^2(\gamma)x^2 + \cos^2(\gamma)y^2 + 2\sin(\gamma)\cos(\gamma)xy + z^2\big)(\partial_x^2 f) \\
& +\big(\cos^2(\gamma)x^2 + \sin^2(\gamma)y^2 - 2\sin(\gamma)\cos(\gamma)xy + z^2\big)\big(\partial_y^2 f\big) \\
& +(x^2+y^2)(\partial_z^2 f) \\
& -2\big((x^2-y^2)\sin(\gamma)\cos(\gamma) + \big(\cos^2(\gamma) - \sin^2(\gamma)\big)xy\big)\big(\partial_x\partial_y f\big) \\
& -2(\cos(\gamma)xz - \sin(\gamma)yz)(\partial_x\partial_z f) - 2(\sin(\gamma)xz + \cos(\gamma)yz)\big(\partial_y\partial_z f\big) \\
& -2(\cos(\gamma)x - \sin(\gamma)y)(\partial_x f) - 2(\sin(\gamma)x + \cos(\gamma)y)\big(\partial_y f\big) - 2z(\partial_z f) \\
\\
= {} & \exp(\gamma J_z)\big(\Delta_{\text{sph}} f\big)
\end{aligned}
$$

For reference, one can go back to equation (5.12) to compare the terms.

So, we conclude that $\Delta_{\text{sph}} g = \Delta\text{sph}(\exp(\gamma J_z)f) = \exp(\gamma J_z)\big(\Delta_{\text{sph}} f\big)$ for all $\gamma \in \mathbb{R}$. The other two elementary rotation matrices also follow similar calculation (here we don't have enough space so we'll skip it). Hence the Spherical Laplace Operator is invariant under $\mathrm{SO}(3, \mathbb{R})$ action (or the rotation action of inputs).